Transforming Growth Factor-Beta Receptor Signaling Drives SMAD2/3 Phosphorylation in Tissue Fibrosis

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Page 1 — Abstract & Background

Transforming growth factor beta (TGF-beta) is an essential cytokine promoting extracellular matrix accumulation and organ fibrosis.

Page 2 — Receptor Activation

TGF-beta binds the TGFBR2 kinase, which recruits and activates TGFBR1 through serine phosphorylation.

Page 3 — SMAD Phosphorylation

Activated TGFBR1 directly phosphorylates SMAD3 at its C-terminal SXS motif, promoting heterotrimer formation with SMAD4.

Page 4 — Results: ECM Synthesis

Nuclear SMAD3 complexes upregulate collagen synthesis and drive alpha-SMA gene expression in myofibroblasts.

Page 5 — Discussion & Conclusions

These findings demonstrate that TGF-beta stimulates collagen production through SMAD3 phosphorylation during progressive fibrosis.